P P SAVANI UNIVERSITY

Seventh Semester of BCA/B.Sc.IT Examination

October 2026

SSCA4030 Image Processing

15.10.2026, Thursday **Time: 12:30 p.m. To 03:00 p.m.** **Maximum Marks: 60**

Instructions:

1. The question paper comprises of two sections.
2. Make suitable assumptions and draw neat figures wherever required.
3. Use of scientific calculator is allowed.

SECTION – I

Q - 1	**MCQ/Short Question/Fill in the Blanks (Any Five)**	**[05]**	**CO**	**BTL**
(i)	A pixel of an 8-bit grayscale image can take values from _____ to _____.		CO1	1
(ii)	How many bits are required to store a binary image of size 64 × 64?		CO1	3
(iii)	Which filter is most suitable for removing salt-and-pepper noise? (a) Mean filter (b) Median filter (c) Laplacian filter (d) High-pass filter		CO2	2
(iv)	The negative of an 8-bit image is obtained using the transformation s = _____.		CO2	1
(v)	Define 4-neighbours $N_4(p)$ of a pixel p located at (x, y).		CO1	1
(vi)	A low-pass filter in the frequency domain is used for _____ of an image.		CO2	1
(vii)	Write the full forms of RGB, CMY and HSI.		CO1	1
Q - 2	**Attempt any two.**	**[10]**		
(i)	Explain sampling and quantization of an image with a neat diagram.		CO1	2
(ii)	Explain fundamentals steps in digital image processing with proper diagram,		CO1	3
(iii)	Differentiate between low pass filter and high pass filter.		CO2	4
(iv)	For the 3 × 3 neighborhood given below, find the new value of the centre pixel using (a) a 3 × 3 mean filter and (b) a 3 × 3 median filter. Which filter gives the better result here? Justify.		CO2	5

10	12	11
13	200	12
11	10	14

Q - 3	**Attempt any two.**							**[10]**	
(i)	Perform histogram equalization on the following 3-bit image (L = 8) of size 8 × 8 and show the new gray levels.								CO2	3

Gray level (r_k)	**0**	**1**	**2**	**3**	**4**	**5**	**6**	**7**
n_k	8	10	10	2	12	16	4	2

(ii)	List and discuss any one basic data redundancy in brief.		CO2	3
(iii)	Explain the RGB and CMY colour models with neat diagrams. Convert the normalized RGB colour (1, 0, 0.5) into the CMY model.		CO1	2
Q - 4	**Attempt any one.**	**[05]**		
(i)	Draw and explain the image degradation/restoration model. Write the basic idea of inverse filtering and Wiener filtering (concept only).		CO2	2
(ii)	Differentiate between Image Enhancement and Image Restoration.		CO2	4

SECTION – II

Q - 1	**MCQ/Short Question/Fill in the Blanks (Any Five)**	**[05]**	**CO**	**BTL**
(i)	Run-Length Encoding (RLE) is a _____ (lossy / lossless) compression technique.		CO3	1

(ii)	JPEG compression is based on the _____ transform.		CO3	1
(iii)	An image of 200 KB is compressed to 50 KB. Find the compression ratio.		CO3	3
(iv)	Dilation followed by erosion is called _____.		CO3	1
(v)	Which edge detector uses non-maximum suppression and hysteresis thresholding? (a) Sobel (b) Prewitt (c) Canny (d) Roberts		CO4	2
(vi)	The Hough transform is mainly used to detect _____ and _____ in an image.		CO4	1
(vii)	Write the full form of OCR.		CO5	1
Q - 2	**Attempt any two.**	**[10]**		
(i)	What is the need for image compression? Explain coding, inter-pixel and psycho-visual redundancy.		CO3	2
(ii)	Differentiate between Sobel and Canny edge detection techniques.		CO3	4
(iii)	Encode the following row of an 8-bit image using Run-Length Encoding (RLE) in (value, count) form. Assuming 1 byte each for value and count, find the compression ratio. 255 255 255 255 255 0 0 0 128 128 128 128 255 255 255 255		CO3	3
(iv)	Differentiate between lossless and lossy compression. Explain the basic steps of JPEG compression with a block diagram.		CO3	2
Q - 3	**Attempt any two.**	**[10]**		
(i)	Define binary morphology and explain erosion and dilation with suitable examples.		CO4	3
(ii)	An image has 100 pixels with six gray levels A to F. Their frequencies are given below. Construct the Huffman code for each symbol, and find the average code length and the compression achieved.		CO3	6

Symbol	**A (White)**	**B**	**C**	**D**	**E**	**F**
Frequency	45	13	12	16	9	5

(iii)	Explain region-based segmentation: (a) Region growing (b) Region splitting and merging.		CO4	2
Q - 4	**Attempt any one.**	**[05]**		
(i)	Describe the applications of morphological image processing.		CO3	3
(ii)	Explain any two applications of image processing from medical imaging, remote sensing and security/biometrics. Write the basic steps of face detection using OpenCV.		CO5	2

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CO : Course Outcome Number BTL : Blooms Taxonomy Level

Level of Bloom's Revised Taxonomy in Assessment

1: Remember	2: Understand	3: Apply
4: Analyze	5: Evaluate	6: Create